

Lab 2 - Combinatorics.docx

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Example

```
Minors = c("Analytics", "Math", "CS", "Stats")
require(DescTools)

## Loading required package: DescTools
## Warning: package 'DescTools' was built under R version 4.4.3
##
## Attaching package: 'DescTools'
## The following object is masked from 'package:mosaic':
##
##      MAD
MinorCombos=CombSet(Minors, 2, repl=F, ord=F)
View(MinorCombos)
head(MinorCombos)

##      [,1]      [,2]
## [1,] "Analytics" "Math"
## [2,] "Analytics" "CS"
## [3,] "Analytics" "Stats"
## [4,] "Math"      "CS"
## [5,] "Math"      "Stats"
## [6,] "CS"        "Stats"
tail(MinorCombos)

##      [,1]      [,2]
## [1,] "Analytics" "Math"
## [2,] "Analytics" "CS"
## [3,] "Analytics" "Stats"
## [4,] "Math"      "CS"
## [5,] "Math"      "Stats"
## [6,] "CS"        "Stats"
nrow(MinorCombos)

## [1] 6
```

Ice Cream

```
set.seed(67)
Flavors <- c("Vanilla", "Chocolate", "Strawberry", "Cookies & Cream", "Cookie Dough", "Mint Chocolate Chip")
```

Scenario 1: Unordered, without replacement (Combinations)

```
Scenario1 <- CombSet(Flavors, 3, repl = FALSE, ord = FALSE)
print(Scenario1) # Print Total
```

##	[,1]	[,2]	[,3]
## [1,]	"Vanilla"	"Chocolate"	"Strawberry"
## [2,]	"Vanilla"	"Chocolate"	"Cookies & Cream"
## [3,]	"Vanilla"	"Chocolate"	"Cookie Dough"
## [4,]	"Vanilla"	"Chocolate"	"Mint Chocolate Chip"
## [5,]	"Vanilla"	"Chocolate"	"Butter Pecan"
## [6,]	"Vanilla"	"Chocolate"	"Coffee"
## [7,]	"Vanilla"	"Strawberry"	"Cookies & Cream"
## [8,]	"Vanilla"	"Strawberry"	"Cookie Dough"
## [9,]	"Vanilla"	"Strawberry"	"Mint Chocolate Chip"
## [10,]	"Vanilla"	"Strawberry"	"Butter Pecan"
## [11,]	"Vanilla"	"Strawberry"	"Coffee"
## [12,]	"Vanilla"	"Cookies & Cream"	"Cookie Dough"
## [13,]	"Vanilla"	"Cookies & Cream"	"Mint Chocolate Chip"
## [14,]	"Vanilla"	"Cookies & Cream"	"Butter Pecan"
## [15,]	"Vanilla"	"Cookies & Cream"	"Coffee"
## [16,]	"Vanilla"	"Cookie Dough"	"Mint Chocolate Chip"
## [17,]	"Vanilla"	"Cookie Dough"	"Butter Pecan"
## [18,]	"Vanilla"	"Cookie Dough"	"Coffee"
## [19,]	"Vanilla"	"Mint Chocolate Chip"	"Butter Pecan"
## [20,]	"Vanilla"	"Mint Chocolate Chip"	"Coffee"
## [21,]	"Vanilla"	"Butter Pecan"	"Coffee"
## [22,]	"Chocolate"	"Strawberry"	"Cookies & Cream"
## [23,]	"Chocolate"	"Strawberry"	"Cookie Dough"
## [24,]	"Chocolate"	"Strawberry"	"Mint Chocolate Chip"
## [25,]	"Chocolate"	"Strawberry"	"Butter Pecan"
## [26,]	"Chocolate"	"Strawberry"	"Coffee"
## [27,]	"Chocolate"	"Cookies & Cream"	"Cookie Dough"
## [28,]	"Chocolate"	"Cookies & Cream"	"Mint Chocolate Chip"
## [29,]	"Chocolate"	"Cookies & Cream"	"Butter Pecan"
## [30,]	"Chocolate"	"Cookies & Cream"	"Coffee"
## [31,]	"Chocolate"	"Cookie Dough"	"Mint Chocolate Chip"
## [32,]	"Chocolate"	"Cookie Dough"	"Butter Pecan"
## [33,]	"Chocolate"	"Cookie Dough"	"Coffee"
## [34,]	"Chocolate"	"Mint Chocolate Chip"	"Butter Pecan"
## [35,]	"Chocolate"	"Mint Chocolate Chip"	"Coffee"
## [36,]	"Chocolate"	"Butter Pecan"	"Coffee"
## [37,]	"Strawberry"	"Cookies & Cream"	"Cookie Dough"
## [38,]	"Strawberry"	"Cookies & Cream"	"Mint Chocolate Chip"
## [39,]	"Strawberry"	"Cookies & Cream"	"Butter Pecan"
## [40,]	"Strawberry"	"Cookies & Cream"	"Coffee"
## [41,]	"Strawberry"	"Cookie Dough"	"Mint Chocolate Chip"
## [42,]	"Strawberry"	"Cookie Dough"	"Butter Pecan"

```
## [43,] "Strawberry"      "Cookie Dough"      "Coffee"
## [44,] "Strawberry"      "Mint Chocolate Chip" "Butter Pecan"
## [45,] "Strawberry"      "Mint Chocolate Chip" "Coffee"
## [46,] "Strawberry"      "Butter Pecan"       "Coffee"
## [47,] "Cookies & Cream" "Cookie Dough"       "Mint Chocolate Chip"
## [48,] "Cookies & Cream" "Cookie Dough"       "Butter Pecan"
## [49,] "Cookies & Cream" "Cookie Dough"       "Coffee"
## [50,] "Cookies & Cream" "Mint Chocolate Chip" "Butter Pecan"
## [51,] "Cookies & Cream" "Mint Chocolate Chip" "Coffee"
## [52,] "Cookies & Cream" "Butter Pecan"       "Coffee"
## [53,] "Cookie Dough"    "Mint Chocolate Chip" "Butter Pecan"
## [54,] "Cookie Dough"    "Mint Chocolate Chip" "Coffee"
## [55,] "Cookie Dough"    "Butter Pecan"       "Coffee"
## [56,] "Mint Chocolate Chip" "Butter Pecan"       "Coffee"
```

```
print(head(Scenario1, 10)) # First 10
```

```
##      [,1]      [,2]      [,3]
## [1,] "Vanilla" "Chocolate" "Strawberry"
## [2,] "Vanilla" "Chocolate" "Cookies & Cream"
## [3,] "Vanilla" "Chocolate" "Cookie Dough"
## [4,] "Vanilla" "Chocolate" "Mint Chocolate Chip"
## [5,] "Vanilla" "Chocolate" "Butter Pecan"
## [6,] "Vanilla" "Chocolate" "Coffee"
## [7,] "Vanilla" "Strawberry" "Cookies & Cream"
## [8,] "Vanilla" "Strawberry" "Cookie Dough"
## [9,] "Vanilla" "Strawberry" "Mint Chocolate Chip"
## [10,] "Vanilla" "Strawberry" "Butter Pecan"
```

```
print(tail(Scenario1, 10)) # Last 10
```

```
##      [,1]      [,2]      [,3]
## [47,] "Cookies & Cream" "Cookie Dough"      "Mint Chocolate Chip"
## [48,] "Cookies & Cream" "Cookie Dough"      "Butter Pecan"
## [49,] "Cookies & Cream" "Cookie Dough"      "Coffee"
## [50,] "Cookies & Cream" "Mint Chocolate Chip" "Butter Pecan"
## [51,] "Cookies & Cream" "Mint Chocolate Chip" "Coffee"
## [52,] "Cookies & Cream" "Butter Pecan"       "Coffee"
## [53,] "Cookie Dough"    "Mint Chocolate Chip" "Butter Pecan"
## [54,] "Cookie Dough"    "Mint Chocolate Chip" "Coffee"
## [55,] "Cookie Dough"    "Butter Pecan"       "Coffee"
## [56,] "Mint Chocolate Chip" "Butter Pecan"       "Coffee"
```

```
print(Scenario1[sample(nrow(Scenario1), 5), ]) # 5 Random
```

```
##      [,1]      [,2]      [,3]
## [1,] "Strawberry" "Cookie Dough"      "Coffee"
## [2,] "Vanilla"    "Mint Chocolate Chip" "Coffee"
## [3,] "Chocolate"  "Cookie Dough"      "Coffee"
## [4,] "Vanilla"    "Cookies & Cream"     "Butter Pecan"
## [5,] "Vanilla"    "Strawberry"         "Cookies & Cream"
```

```
#56 total combos in first print
```

```
print(choose(8, 3)) #8c3 compare
```

```
## [1] 56
```

```
#Same
```

Scenario 2: Ordered, without replacement (Permutations)

```
Scenario2 <- CombSet(Flavors, 3, repl = FALSE, ord = TRUE)
print(Scenario2) # Print Total
```

##	[,1]	[,2]	[,3]
##	[1,] "Vanilla"	"Chocolate"	"Strawberry"
##	[2,] "Chocolate"	"Vanilla"	"Strawberry"
##	[3,] "Chocolate"	"Strawberry"	"Vanilla"
##	[4,] "Vanilla"	"Strawberry"	"Chocolate"
##	[5,] "Strawberry"	"Vanilla"	"Chocolate"
##	[6,] "Strawberry"	"Chocolate"	"Vanilla"
##	[7,] "Vanilla"	"Chocolate"	"Cookies & Cream"
##	[8,] "Chocolate"	"Vanilla"	"Cookies & Cream"
##	[9,] "Chocolate"	"Cookies & Cream"	"Vanilla"
##	[10,] "Vanilla"	"Cookies & Cream"	"Chocolate"
##	[11,] "Cookies & Cream"	"Vanilla"	"Chocolate"
##	[12,] "Cookies & Cream"	"Chocolate"	"Vanilla"
##	[13,] "Vanilla"	"Chocolate"	"Cookie Dough"
##	[14,] "Chocolate"	"Vanilla"	"Cookie Dough"
##	[15,] "Chocolate"	"Cookie Dough"	"Vanilla"
##	[16,] "Vanilla"	"Cookie Dough"	"Chocolate"
##	[17,] "Cookie Dough"	"Vanilla"	"Chocolate"
##	[18,] "Cookie Dough"	"Chocolate"	"Vanilla"
##	[19,] "Vanilla"	"Chocolate"	"Mint Chocolate Chip"
##	[20,] "Chocolate"	"Vanilla"	"Mint Chocolate Chip"
##	[21,] "Chocolate"	"Mint Chocolate Chip"	"Vanilla"
##	[22,] "Vanilla"	"Mint Chocolate Chip"	"Chocolate"
##	[23,] "Mint Chocolate Chip"	"Vanilla"	"Chocolate"
##	[24,] "Mint Chocolate Chip"	"Chocolate"	"Vanilla"
##	[25,] "Vanilla"	"Chocolate"	"Butter Pecan"
##	[26,] "Chocolate"	"Vanilla"	"Butter Pecan"
##	[27,] "Chocolate"	"Butter Pecan"	"Vanilla"
##	[28,] "Vanilla"	"Butter Pecan"	"Chocolate"
##	[29,] "Butter Pecan"	"Vanilla"	"Chocolate"
##	[30,] "Butter Pecan"	"Chocolate"	"Vanilla"
##	[31,] "Vanilla"	"Chocolate"	"Coffee"
##	[32,] "Chocolate"	"Vanilla"	"Coffee"
##	[33,] "Chocolate"	"Coffee"	"Vanilla"
##	[34,] "Vanilla"	"Coffee"	"Chocolate"
##	[35,] "Coffee"	"Vanilla"	"Chocolate"
##	[36,] "Coffee"	"Chocolate"	"Vanilla"
##	[37,] "Vanilla"	"Strawberry"	"Cookies & Cream"
##	[38,] "Strawberry"	"Vanilla"	"Cookies & Cream"
##	[39,] "Strawberry"	"Cookies & Cream"	"Vanilla"
##	[40,] "Vanilla"	"Cookies & Cream"	"Strawberry"
##	[41,] "Cookies & Cream"	"Vanilla"	"Strawberry"
##	[42,] "Cookies & Cream"	"Strawberry"	"Vanilla"
##	[43,] "Vanilla"	"Strawberry"	"Cookie Dough"
##	[44,] "Strawberry"	"Vanilla"	"Cookie Dough"
##	[45,] "Strawberry"	"Cookie Dough"	"Vanilla"

##	[46,]	"Vanilla"	"Cookie Dough"	"Strawberry"
##	[47,]	"Cookie Dough"	"Vanilla"	"Strawberry"
##	[48,]	"Cookie Dough"	"Strawberry"	"Vanilla"
##	[49,]	"Vanilla"	"Strawberry"	"Mint Chocolate Chip"
##	[50,]	"Strawberry"	"Vanilla"	"Mint Chocolate Chip"
##	[51,]	"Strawberry"	"Mint Chocolate Chip"	"Vanilla"
##	[52,]	"Vanilla"	"Mint Chocolate Chip"	"Strawberry"
##	[53,]	"Mint Chocolate Chip"	"Vanilla"	"Strawberry"
##	[54,]	"Mint Chocolate Chip"	"Strawberry"	"Vanilla"
##	[55,]	"Vanilla"	"Strawberry"	"Butter Pecan"
##	[56,]	"Strawberry"	"Vanilla"	"Butter Pecan"
##	[57,]	"Strawberry"	"Butter Pecan"	"Vanilla"
##	[58,]	"Vanilla"	"Butter Pecan"	"Strawberry"
##	[59,]	"Butter Pecan"	"Vanilla"	"Strawberry"
##	[60,]	"Butter Pecan"	"Strawberry"	"Vanilla"
##	[61,]	"Vanilla"	"Strawberry"	"Coffee"
##	[62,]	"Strawberry"	"Vanilla"	"Coffee"
##	[63,]	"Strawberry"	"Coffee"	"Vanilla"
##	[64,]	"Vanilla"	"Coffee"	"Strawberry"
##	[65,]	"Coffee"	"Vanilla"	"Strawberry"
##	[66,]	"Coffee"	"Strawberry"	"Vanilla"
##	[67,]	"Vanilla"	"Cookies & Cream"	"Cookie Dough"
##	[68,]	"Cookies & Cream"	"Vanilla"	"Cookie Dough"
##	[69,]	"Cookies & Cream"	"Cookie Dough"	"Vanilla"
##	[70,]	"Vanilla"	"Cookie Dough"	"Cookies & Cream"
##	[71,]	"Cookie Dough"	"Vanilla"	"Cookies & Cream"
##	[72,]	"Cookie Dough"	"Cookies & Cream"	"Vanilla"
##	[73,]	"Vanilla"	"Cookies & Cream"	"Mint Chocolate Chip"
##	[74,]	"Cookies & Cream"	"Vanilla"	"Mint Chocolate Chip"
##	[75,]	"Cookies & Cream"	"Mint Chocolate Chip"	"Vanilla"
##	[76,]	"Vanilla"	"Mint Chocolate Chip"	"Cookies & Cream"
##	[77,]	"Mint Chocolate Chip"	"Vanilla"	"Cookies & Cream"
##	[78,]	"Mint Chocolate Chip"	"Cookies & Cream"	"Vanilla"
##	[79,]	"Vanilla"	"Cookies & Cream"	"Butter Pecan"
##	[80,]	"Cookies & Cream"	"Vanilla"	"Butter Pecan"
##	[81,]	"Cookies & Cream"	"Butter Pecan"	"Vanilla"
##	[82,]	"Vanilla"	"Butter Pecan"	"Cookies & Cream"
##	[83,]	"Butter Pecan"	"Vanilla"	"Cookies & Cream"
##	[84,]	"Butter Pecan"	"Cookies & Cream"	"Vanilla"
##	[85,]	"Vanilla"	"Cookies & Cream"	"Coffee"
##	[86,]	"Cookies & Cream"	"Vanilla"	"Coffee"
##	[87,]	"Cookies & Cream"	"Coffee"	"Vanilla"
##	[88,]	"Vanilla"	"Coffee"	"Cookies & Cream"
##	[89,]	"Coffee"	"Vanilla"	"Cookies & Cream"
##	[90,]	"Coffee"	"Cookies & Cream"	"Vanilla"
##	[91,]	"Vanilla"	"Cookie Dough"	"Mint Chocolate Chip"
##	[92,]	"Cookie Dough"	"Vanilla"	"Mint Chocolate Chip"
##	[93,]	"Cookie Dough"	"Mint Chocolate Chip"	"Vanilla"
##	[94,]	"Vanilla"	"Mint Chocolate Chip"	"Cookie Dough"
##	[95,]	"Mint Chocolate Chip"	"Vanilla"	"Cookie Dough"
##	[96,]	"Mint Chocolate Chip"	"Cookie Dough"	"Vanilla"
##	[97,]	"Vanilla"	"Cookie Dough"	"Butter Pecan"
##	[98,]	"Cookie Dough"	"Vanilla"	"Butter Pecan"
##	[99,]	"Cookie Dough"	"Butter Pecan"	"Vanilla"

## [100,]	"Vanilla"	"Butter Pecan"	"Cookie Dough"
## [101,]	"Butter Pecan"	"Vanilla"	"Cookie Dough"
## [102,]	"Butter Pecan"	"Cookie Dough"	"Vanilla"
## [103,]	"Vanilla"	"Cookie Dough"	"Coffee"
## [104,]	"Cookie Dough"	"Vanilla"	"Coffee"
## [105,]	"Cookie Dough"	"Coffee"	"Vanilla"
## [106,]	"Vanilla"	"Coffee"	"Cookie Dough"
## [107,]	"Coffee"	"Vanilla"	"Cookie Dough"
## [108,]	"Coffee"	"Cookie Dough"	"Vanilla"
## [109,]	"Vanilla"	"Mint Chocolate Chip"	"Butter Pecan"
## [110,]	"Mint Chocolate Chip"	"Vanilla"	"Butter Pecan"
## [111,]	"Mint Chocolate Chip"	"Butter Pecan"	"Vanilla"
## [112,]	"Vanilla"	"Butter Pecan"	"Mint Chocolate Chip"
## [113,]	"Butter Pecan"	"Vanilla"	"Mint Chocolate Chip"
## [114,]	"Butter Pecan"	"Mint Chocolate Chip"	"Vanilla"
## [115,]	"Vanilla"	"Mint Chocolate Chip"	"Coffee"
## [116,]	"Mint Chocolate Chip"	"Vanilla"	"Coffee"
## [117,]	"Mint Chocolate Chip"	"Coffee"	"Vanilla"
## [118,]	"Vanilla"	"Coffee"	"Mint Chocolate Chip"
## [119,]	"Coffee"	"Vanilla"	"Mint Chocolate Chip"
## [120,]	"Coffee"	"Mint Chocolate Chip"	"Vanilla"
## [121,]	"Vanilla"	"Butter Pecan"	"Coffee"
## [122,]	"Butter Pecan"	"Vanilla"	"Coffee"
## [123,]	"Butter Pecan"	"Coffee"	"Vanilla"
## [124,]	"Vanilla"	"Coffee"	"Butter Pecan"
## [125,]	"Coffee"	"Vanilla"	"Butter Pecan"
## [126,]	"Coffee"	"Butter Pecan"	"Vanilla"
## [127,]	"Chocolate"	"Strawberry"	"Cookies & Cream"
## [128,]	"Strawberry"	"Chocolate"	"Cookies & Cream"
## [129,]	"Strawberry"	"Cookies & Cream"	"Chocolate"
## [130,]	"Chocolate"	"Cookies & Cream"	"Strawberry"
## [131,]	"Cookies & Cream"	"Chocolate"	"Strawberry"
## [132,]	"Cookies & Cream"	"Strawberry"	"Chocolate"
## [133,]	"Chocolate"	"Strawberry"	"Cookie Dough"
## [134,]	"Strawberry"	"Chocolate"	"Cookie Dough"
## [135,]	"Strawberry"	"Cookie Dough"	"Chocolate"
## [136,]	"Chocolate"	"Cookie Dough"	"Strawberry"
## [137,]	"Cookie Dough"	"Chocolate"	"Strawberry"
## [138,]	"Cookie Dough"	"Strawberry"	"Chocolate"
## [139,]	"Chocolate"	"Strawberry"	"Mint Chocolate Chip"
## [140,]	"Strawberry"	"Chocolate"	"Mint Chocolate Chip"
## [141,]	"Strawberry"	"Mint Chocolate Chip"	"Chocolate"
## [142,]	"Chocolate"	"Mint Chocolate Chip"	"Strawberry"
## [143,]	"Mint Chocolate Chip"	"Chocolate"	"Strawberry"
## [144,]	"Mint Chocolate Chip"	"Strawberry"	"Chocolate"
## [145,]	"Chocolate"	"Strawberry"	"Butter Pecan"
## [146,]	"Strawberry"	"Chocolate"	"Butter Pecan"
## [147,]	"Strawberry"	"Butter Pecan"	"Chocolate"
## [148,]	"Chocolate"	"Butter Pecan"	"Strawberry"
## [149,]	"Butter Pecan"	"Chocolate"	"Strawberry"
## [150,]	"Butter Pecan"	"Strawberry"	"Chocolate"
## [151,]	"Chocolate"	"Strawberry"	"Coffee"
## [152,]	"Strawberry"	"Chocolate"	"Coffee"
## [153,]	"Strawberry"	"Coffee"	"Chocolate"

## [154,]	"Chocolate"	"Coffee"	"Strawberry"
## [155,]	"Coffee"	"Chocolate"	"Strawberry"
## [156,]	"Coffee"	"Strawberry"	"Chocolate"
## [157,]	"Chocolate"	"Cookies & Cream"	"Cookie Dough"
## [158,]	"Cookies & Cream"	"Chocolate"	"Cookie Dough"
## [159,]	"Cookies & Cream"	"Cookie Dough"	"Chocolate"
## [160,]	"Chocolate"	"Cookie Dough"	"Cookies & Cream"
## [161,]	"Cookie Dough"	"Chocolate"	"Cookies & Cream"
## [162,]	"Cookie Dough"	"Cookies & Cream"	"Chocolate"
## [163,]	"Chocolate"	"Cookies & Cream"	"Mint Chocolate Chip"
## [164,]	"Cookies & Cream"	"Chocolate"	"Mint Chocolate Chip"
## [165,]	"Cookies & Cream"	"Mint Chocolate Chip"	"Chocolate"
## [166,]	"Chocolate"	"Mint Chocolate Chip"	"Cookies & Cream"
## [167,]	"Mint Chocolate Chip"	"Chocolate"	"Cookies & Cream"
## [168,]	"Mint Chocolate Chip"	"Cookies & Cream"	"Chocolate"
## [169,]	"Chocolate"	"Cookies & Cream"	"Butter Pecan"
## [170,]	"Cookies & Cream"	"Chocolate"	"Butter Pecan"
## [171,]	"Cookies & Cream"	"Butter Pecan"	"Chocolate"
## [172,]	"Chocolate"	"Butter Pecan"	"Cookies & Cream"
## [173,]	"Butter Pecan"	"Chocolate"	"Cookies & Cream"
## [174,]	"Butter Pecan"	"Cookies & Cream"	"Chocolate"
## [175,]	"Chocolate"	"Cookies & Cream"	"Coffee"
## [176,]	"Cookies & Cream"	"Chocolate"	"Coffee"
## [177,]	"Cookies & Cream"	"Coffee"	"Chocolate"
## [178,]	"Chocolate"	"Coffee"	"Cookies & Cream"
## [179,]	"Coffee"	"Chocolate"	"Cookies & Cream"
## [180,]	"Coffee"	"Cookies & Cream"	"Chocolate"
## [181,]	"Chocolate"	"Cookie Dough"	"Mint Chocolate Chip"
## [182,]	"Cookie Dough"	"Chocolate"	"Mint Chocolate Chip"
## [183,]	"Cookie Dough"	"Mint Chocolate Chip"	"Chocolate"
## [184,]	"Chocolate"	"Mint Chocolate Chip"	"Cookie Dough"
## [185,]	"Mint Chocolate Chip"	"Chocolate"	"Cookie Dough"
## [186,]	"Mint Chocolate Chip"	"Cookie Dough"	"Chocolate"
## [187,]	"Chocolate"	"Cookie Dough"	"Butter Pecan"
## [188,]	"Cookie Dough"	"Chocolate"	"Butter Pecan"
## [189,]	"Cookie Dough"	"Butter Pecan"	"Chocolate"
## [190,]	"Chocolate"	"Butter Pecan"	"Cookie Dough"
## [191,]	"Butter Pecan"	"Chocolate"	"Cookie Dough"
## [192,]	"Butter Pecan"	"Cookie Dough"	"Chocolate"
## [193,]	"Chocolate"	"Cookie Dough"	"Coffee"
## [194,]	"Cookie Dough"	"Chocolate"	"Coffee"
## [195,]	"Cookie Dough"	"Coffee"	"Chocolate"
## [196,]	"Chocolate"	"Coffee"	"Cookie Dough"
## [197,]	"Coffee"	"Chocolate"	"Cookie Dough"
## [198,]	"Coffee"	"Cookie Dough"	"Chocolate"
## [199,]	"Chocolate"	"Mint Chocolate Chip"	"Butter Pecan"
## [200,]	"Mint Chocolate Chip"	"Chocolate"	"Butter Pecan"
## [201,]	"Mint Chocolate Chip"	"Butter Pecan"	"Chocolate"
## [202,]	"Chocolate"	"Butter Pecan"	"Mint Chocolate Chip"
## [203,]	"Butter Pecan"	"Chocolate"	"Mint Chocolate Chip"
## [204,]	"Butter Pecan"	"Mint Chocolate Chip"	"Chocolate"
## [205,]	"Chocolate"	"Mint Chocolate Chip"	"Coffee"
## [206,]	"Mint Chocolate Chip"	"Chocolate"	"Coffee"
## [207,]	"Mint Chocolate Chip"	"Coffee"	"Chocolate"

## [208,]	"Chocolate"	"Coffee"	"Mint Chocolate Chip"
## [209,]	"Coffee"	"Chocolate"	"Mint Chocolate Chip"
## [210,]	"Coffee"	"Mint Chocolate Chip"	"Chocolate"
## [211,]	"Chocolate"	"Butter Pecan"	"Coffee"
## [212,]	"Butter Pecan"	"Chocolate"	"Coffee"
## [213,]	"Butter Pecan"	"Coffee"	"Chocolate"
## [214,]	"Chocolate"	"Coffee"	"Butter Pecan"
## [215,]	"Coffee"	"Chocolate"	"Butter Pecan"
## [216,]	"Coffee"	"Butter Pecan"	"Chocolate"
## [217,]	"Strawberry"	"Cookies & Cream"	"Cookie Dough"
## [218,]	"Cookies & Cream"	"Strawberry"	"Cookie Dough"
## [219,]	"Cookies & Cream"	"Cookie Dough"	"Strawberry"
## [220,]	"Strawberry"	"Cookie Dough"	"Cookies & Cream"
## [221,]	"Cookie Dough"	"Strawberry"	"Cookies & Cream"
## [222,]	"Cookie Dough"	"Cookies & Cream"	"Strawberry"
## [223,]	"Strawberry"	"Cookies & Cream"	"Mint Chocolate Chip"
## [224,]	"Cookies & Cream"	"Strawberry"	"Mint Chocolate Chip"
## [225,]	"Cookies & Cream"	"Mint Chocolate Chip"	"Strawberry"
## [226,]	"Strawberry"	"Mint Chocolate Chip"	"Cookies & Cream"
## [227,]	"Mint Chocolate Chip"	"Strawberry"	"Cookies & Cream"
## [228,]	"Mint Chocolate Chip"	"Cookies & Cream"	"Strawberry"
## [229,]	"Strawberry"	"Cookies & Cream"	"Butter Pecan"
## [230,]	"Cookies & Cream"	"Strawberry"	"Butter Pecan"
## [231,]	"Cookies & Cream"	"Butter Pecan"	"Strawberry"
## [232,]	"Strawberry"	"Butter Pecan"	"Cookies & Cream"
## [233,]	"Butter Pecan"	"Strawberry"	"Cookies & Cream"
## [234,]	"Butter Pecan"	"Cookies & Cream"	"Strawberry"
## [235,]	"Strawberry"	"Cookies & Cream"	"Coffee"
## [236,]	"Cookies & Cream"	"Strawberry"	"Coffee"
## [237,]	"Cookies & Cream"	"Coffee"	"Strawberry"
## [238,]	"Strawberry"	"Coffee"	"Cookies & Cream"
## [239,]	"Coffee"	"Strawberry"	"Cookies & Cream"
## [240,]	"Coffee"	"Cookies & Cream"	"Strawberry"
## [241,]	"Strawberry"	"Cookie Dough"	"Mint Chocolate Chip"
## [242,]	"Cookie Dough"	"Strawberry"	"Mint Chocolate Chip"
## [243,]	"Cookie Dough"	"Mint Chocolate Chip"	"Strawberry"
## [244,]	"Strawberry"	"Mint Chocolate Chip"	"Cookie Dough"
## [245,]	"Mint Chocolate Chip"	"Strawberry"	"Cookie Dough"
## [246,]	"Mint Chocolate Chip"	"Cookie Dough"	"Strawberry"
## [247,]	"Strawberry"	"Cookie Dough"	"Butter Pecan"
## [248,]	"Cookie Dough"	"Strawberry"	"Butter Pecan"
## [249,]	"Cookie Dough"	"Butter Pecan"	"Strawberry"
## [250,]	"Strawberry"	"Butter Pecan"	"Cookie Dough"
## [251,]	"Butter Pecan"	"Strawberry"	"Cookie Dough"
## [252,]	"Butter Pecan"	"Cookie Dough"	"Strawberry"
## [253,]	"Strawberry"	"Cookie Dough"	"Coffee"
## [254,]	"Cookie Dough"	"Strawberry"	"Coffee"
## [255,]	"Cookie Dough"	"Coffee"	"Strawberry"
## [256,]	"Strawberry"	"Coffee"	"Cookie Dough"
## [257,]	"Coffee"	"Strawberry"	"Cookie Dough"
## [258,]	"Coffee"	"Cookie Dough"	"Strawberry"
## [259,]	"Strawberry"	"Mint Chocolate Chip"	"Butter Pecan"
## [260,]	"Mint Chocolate Chip"	"Strawberry"	"Butter Pecan"
## [261,]	"Mint Chocolate Chip"	"Butter Pecan"	"Strawberry"

## [262,]	"Strawberry"	"Butter Pecan"	"Mint Chocolate Chip"
## [263,]	"Butter Pecan"	"Strawberry"	"Mint Chocolate Chip"
## [264,]	"Butter Pecan"	"Mint Chocolate Chip"	"Strawberry"
## [265,]	"Strawberry"	"Mint Chocolate Chip"	"Coffee"
## [266,]	"Mint Chocolate Chip"	"Strawberry"	"Coffee"
## [267,]	"Mint Chocolate Chip"	"Coffee"	"Strawberry"
## [268,]	"Strawberry"	"Coffee"	"Mint Chocolate Chip"
## [269,]	"Coffee"	"Strawberry"	"Mint Chocolate Chip"
## [270,]	"Coffee"	"Mint Chocolate Chip"	"Strawberry"
## [271,]	"Strawberry"	"Butter Pecan"	"Coffee"
## [272,]	"Butter Pecan"	"Strawberry"	"Coffee"
## [273,]	"Butter Pecan"	"Coffee"	"Strawberry"
## [274,]	"Strawberry"	"Coffee"	"Butter Pecan"
## [275,]	"Coffee"	"Strawberry"	"Butter Pecan"
## [276,]	"Coffee"	"Butter Pecan"	"Strawberry"
## [277,]	"Cookies & Cream"	"Cookie Dough"	"Mint Chocolate Chip"
## [278,]	"Cookie Dough"	"Cookies & Cream"	"Mint Chocolate Chip"
## [279,]	"Cookie Dough"	"Mint Chocolate Chip"	"Cookies & Cream"
## [280,]	"Cookies & Cream"	"Mint Chocolate Chip"	"Cookie Dough"
## [281,]	"Mint Chocolate Chip"	"Cookies & Cream"	"Cookie Dough"
## [282,]	"Mint Chocolate Chip"	"Cookie Dough"	"Cookies & Cream"
## [283,]	"Cookies & Cream"	"Cookie Dough"	"Butter Pecan"
## [284,]	"Cookie Dough"	"Cookies & Cream"	"Butter Pecan"
## [285,]	"Cookie Dough"	"Butter Pecan"	"Cookies & Cream"
## [286,]	"Cookies & Cream"	"Butter Pecan"	"Cookie Dough"
## [287,]	"Butter Pecan"	"Cookies & Cream"	"Cookie Dough"
## [288,]	"Butter Pecan"	"Cookie Dough"	"Cookies & Cream"
## [289,]	"Cookies & Cream"	"Cookie Dough"	"Coffee"
## [290,]	"Cookie Dough"	"Cookies & Cream"	"Coffee"
## [291,]	"Cookie Dough"	"Coffee"	"Cookies & Cream"
## [292,]	"Cookies & Cream"	"Coffee"	"Cookie Dough"
## [293,]	"Coffee"	"Cookies & Cream"	"Cookie Dough"
## [294,]	"Coffee"	"Cookie Dough"	"Cookies & Cream"
## [295,]	"Cookies & Cream"	"Mint Chocolate Chip"	"Butter Pecan"
## [296,]	"Mint Chocolate Chip"	"Cookies & Cream"	"Butter Pecan"
## [297,]	"Mint Chocolate Chip"	"Butter Pecan"	"Cookies & Cream"
## [298,]	"Cookies & Cream"	"Butter Pecan"	"Mint Chocolate Chip"
## [299,]	"Butter Pecan"	"Cookies & Cream"	"Mint Chocolate Chip"
## [300,]	"Butter Pecan"	"Mint Chocolate Chip"	"Cookies & Cream"
## [301,]	"Cookies & Cream"	"Mint Chocolate Chip"	"Coffee"
## [302,]	"Mint Chocolate Chip"	"Cookies & Cream"	"Coffee"
## [303,]	"Mint Chocolate Chip"	"Coffee"	"Cookies & Cream"
## [304,]	"Cookies & Cream"	"Coffee"	"Mint Chocolate Chip"
## [305,]	"Coffee"	"Cookies & Cream"	"Mint Chocolate Chip"
## [306,]	"Coffee"	"Mint Chocolate Chip"	"Cookies & Cream"
## [307,]	"Cookies & Cream"	"Butter Pecan"	"Coffee"
## [308,]	"Butter Pecan"	"Cookies & Cream"	"Coffee"
## [309,]	"Butter Pecan"	"Coffee"	"Cookies & Cream"
## [310,]	"Cookies & Cream"	"Coffee"	"Butter Pecan"
## [311,]	"Coffee"	"Cookies & Cream"	"Butter Pecan"
## [312,]	"Coffee"	"Butter Pecan"	"Cookies & Cream"
## [313,]	"Cookie Dough"	"Mint Chocolate Chip"	"Butter Pecan"
## [314,]	"Mint Chocolate Chip"	"Cookie Dough"	"Butter Pecan"
## [315,]	"Mint Chocolate Chip"	"Butter Pecan"	"Cookie Dough"

```
## [316,] "Cookie Dough"      "Butter Pecan"      "Mint Chocolate Chip"
## [317,] "Butter Pecan"      "Cookie Dough"      "Mint Chocolate Chip"
## [318,] "Butter Pecan"      "Mint Chocolate Chip" "Cookie Dough"
## [319,] "Cookie Dough"      "Mint Chocolate Chip" "Coffee"
## [320,] "Mint Chocolate Chip" "Cookie Dough"      "Coffee"
## [321,] "Mint Chocolate Chip" "Coffee"            "Cookie Dough"
## [322,] "Cookie Dough"      "Coffee"            "Mint Chocolate Chip"
## [323,] "Coffee"            "Cookie Dough"      "Mint Chocolate Chip"
## [324,] "Coffee"            "Mint Chocolate Chip" "Cookie Dough"
## [325,] "Cookie Dough"      "Butter Pecan"      "Coffee"
## [326,] "Butter Pecan"      "Cookie Dough"      "Coffee"
## [327,] "Butter Pecan"      "Coffee"            "Cookie Dough"
## [328,] "Cookie Dough"      "Coffee"            "Butter Pecan"
## [329,] "Coffee"            "Cookie Dough"      "Butter Pecan"
## [330,] "Coffee"            "Butter Pecan"      "Cookie Dough"
## [331,] "Mint Chocolate Chip" "Butter Pecan"      "Coffee"
## [332,] "Butter Pecan"      "Mint Chocolate Chip" "Coffee"
## [333,] "Butter Pecan"      "Coffee"            "Mint Chocolate Chip"
## [334,] "Mint Chocolate Chip" "Coffee"            "Butter Pecan"
## [335,] "Coffee"            "Mint Chocolate Chip" "Butter Pecan"
## [336,] "Coffee"            "Butter Pecan"      "Mint Chocolate Chip"
```

```
print(head(Scenario2, 10)) # First 10
```

```
##      [,1]      [,2]      [,3]
## [1,] "Vanilla"  "Chocolate" "Strawberry"
## [2,] "Chocolate" "Vanilla"    "Strawberry"
## [3,] "Chocolate" "Strawberry" "Vanilla"
## [4,] "Vanilla"  "Strawberry" "Chocolate"
## [5,] "Strawberry" "Vanilla"    "Chocolate"
## [6,] "Strawberry" "Chocolate"  "Vanilla"
## [7,] "Vanilla"  "Chocolate"  "Cookies & Cream"
## [8,] "Chocolate" "Vanilla"    "Cookies & Cream"
## [9,] "Chocolate" "Cookies & Cream" "Vanilla"
## [10,] "Vanilla"  "Cookies & Cream" "Chocolate"
```

```
print(tail(Scenario2, 10)) # Last 10
```

```
##      [,1]      [,2]      [,3]
## [327,] "Butter Pecan"      "Coffee"      "Cookie Dough"
## [328,] "Cookie Dough"      "Coffee"      "Butter Pecan"
## [329,] "Coffee"            "Cookie Dough" "Butter Pecan"
## [330,] "Coffee"            "Butter Pecan" "Cookie Dough"
## [331,] "Mint Chocolate Chip" "Butter Pecan" "Coffee"
## [332,] "Butter Pecan"      "Mint Chocolate Chip" "Coffee"
## [333,] "Butter Pecan"      "Coffee"        "Mint Chocolate Chip"
## [334,] "Mint Chocolate Chip" "Coffee"        "Butter Pecan"
## [335,] "Coffee"            "Mint Chocolate Chip" "Butter Pecan"
## [336,] "Coffee"            "Butter Pecan"  "Mint Chocolate Chip"
```

```
print(Scenario2[sample(nrow(Scenario2), 5), ]) # 5 Random
```

```
##      [,1]      [,2]      [,3]
## [1,] "Chocolate" "Coffee"      "Vanilla"
## [2,] "Vanilla"   "Mint Chocolate Chip" "Cookie Dough"
## [3,] "Strawberry" "Cookie Dough" "Mint Chocolate Chip"
```

```
## [4,] "Cookies & Cream" "Mint Chocolate Chip" "Coffee"
## [5,] "Vanilla"          "Mint Chocolate Chip" "Strawberry"
```

```
#336 total combos in first print
print(factorial(8) / factorial(5)) #8!/5!=336
```

```
## [1] 336
```

```
#Same
```

Scenario 3: Ordered, with replacement

```
Scenario3 <- CombSet(Flavors, 3, repl = TRUE, ord = TRUE)
print(Scenario3) # Print Total
```

```
##      [,1]      [,2]      [,3]
## [1,] "Vanilla"  "Vanilla"  "Vanilla"
## [2,] "Chocolate" "Vanilla"  "Vanilla"
## [3,] "Strawberry" "Vanilla"  "Vanilla"
## [4,] "Cookies & Cream" "Vanilla"  "Vanilla"
## [5,] "Cookie Dough" "Vanilla"  "Vanilla"
## [6,] "Mint Chocolate Chip" "Vanilla"  "Vanilla"
## [7,] "Butter Pecan" "Vanilla"  "Vanilla"
## [8,] "Coffee" "Vanilla"  "Vanilla"
## [9,] "Vanilla" "Chocolate" "Vanilla"
## [10,] "Chocolate" "Chocolate" "Vanilla"
## [11,] "Strawberry" "Chocolate" "Vanilla"
## [12,] "Cookies & Cream" "Chocolate" "Vanilla"
## [13,] "Cookie Dough" "Chocolate" "Vanilla"
## [14,] "Mint Chocolate Chip" "Chocolate" "Vanilla"
## [15,] "Butter Pecan" "Chocolate" "Vanilla"
## [16,] "Coffee" "Chocolate" "Vanilla"
## [17,] "Vanilla" "Strawberry" "Vanilla"
## [18,] "Chocolate" "Strawberry" "Vanilla"
## [19,] "Strawberry" "Strawberry" "Vanilla"
## [20,] "Cookies & Cream" "Strawberry" "Vanilla"
## [21,] "Cookie Dough" "Strawberry" "Vanilla"
## [22,] "Mint Chocolate Chip" "Strawberry" "Vanilla"
## [23,] "Butter Pecan" "Strawberry" "Vanilla"
## [24,] "Coffee" "Strawberry" "Vanilla"
## [25,] "Vanilla" "Cookies & Cream" "Vanilla"
## [26,] "Chocolate" "Cookies & Cream" "Vanilla"
## [27,] "Strawberry" "Cookies & Cream" "Vanilla"
## [28,] "Cookies & Cream" "Cookies & Cream" "Vanilla"
## [29,] "Cookie Dough" "Cookies & Cream" "Vanilla"
## [30,] "Mint Chocolate Chip" "Cookies & Cream" "Vanilla"
## [31,] "Butter Pecan" "Cookies & Cream" "Vanilla"
## [32,] "Coffee" "Cookies & Cream" "Vanilla"
## [33,] "Vanilla" "Cookie Dough" "Vanilla"
## [34,] "Chocolate" "Cookie Dough" "Vanilla"
## [35,] "Strawberry" "Cookie Dough" "Vanilla"
## [36,] "Cookies & Cream" "Cookie Dough" "Vanilla"
## [37,] "Cookie Dough" "Cookie Dough" "Vanilla"
## [38,] "Mint Chocolate Chip" "Cookie Dough" "Vanilla"
```

##	[39,]	"Butter Pecan"	"Cookie Dough"	"Vanilla"
##	[40,]	"Coffee"	"Cookie Dough"	"Vanilla"
##	[41,]	"Vanilla"	"Mint Chocolate Chip"	"Vanilla"
##	[42,]	"Chocolate"	"Mint Chocolate Chip"	"Vanilla"
##	[43,]	"Strawberry"	"Mint Chocolate Chip"	"Vanilla"
##	[44,]	"Cookies & Cream"	"Mint Chocolate Chip"	"Vanilla"
##	[45,]	"Cookie Dough"	"Mint Chocolate Chip"	"Vanilla"
##	[46,]	"Mint Chocolate Chip"	"Mint Chocolate Chip"	"Vanilla"
##	[47,]	"Butter Pecan"	"Mint Chocolate Chip"	"Vanilla"
##	[48,]	"Coffee"	"Mint Chocolate Chip"	"Vanilla"
##	[49,]	"Vanilla"	"Butter Pecan"	"Vanilla"
##	[50,]	"Chocolate"	"Butter Pecan"	"Vanilla"
##	[51,]	"Strawberry"	"Butter Pecan"	"Vanilla"
##	[52,]	"Cookies & Cream"	"Butter Pecan"	"Vanilla"
##	[53,]	"Cookie Dough"	"Butter Pecan"	"Vanilla"
##	[54,]	"Mint Chocolate Chip"	"Butter Pecan"	"Vanilla"
##	[55,]	"Butter Pecan"	"Butter Pecan"	"Vanilla"
##	[56,]	"Coffee"	"Butter Pecan"	"Vanilla"
##	[57,]	"Vanilla"	"Coffee"	"Vanilla"
##	[58,]	"Chocolate"	"Coffee"	"Vanilla"
##	[59,]	"Strawberry"	"Coffee"	"Vanilla"
##	[60,]	"Cookies & Cream"	"Coffee"	"Vanilla"
##	[61,]	"Cookie Dough"	"Coffee"	"Vanilla"
##	[62,]	"Mint Chocolate Chip"	"Coffee"	"Vanilla"
##	[63,]	"Butter Pecan"	"Coffee"	"Vanilla"
##	[64,]	"Coffee"	"Coffee"	"Vanilla"
##	[65,]	"Vanilla"	"Vanilla"	"Chocolate"
##	[66,]	"Chocolate"	"Vanilla"	"Chocolate"
##	[67,]	"Strawberry"	"Vanilla"	"Chocolate"
##	[68,]	"Cookies & Cream"	"Vanilla"	"Chocolate"
##	[69,]	"Cookie Dough"	"Vanilla"	"Chocolate"
##	[70,]	"Mint Chocolate Chip"	"Vanilla"	"Chocolate"
##	[71,]	"Butter Pecan"	"Vanilla"	"Chocolate"
##	[72,]	"Coffee"	"Vanilla"	"Chocolate"
##	[73,]	"Vanilla"	"Chocolate"	"Chocolate"
##	[74,]	"Chocolate"	"Chocolate"	"Chocolate"
##	[75,]	"Strawberry"	"Chocolate"	"Chocolate"
##	[76,]	"Cookies & Cream"	"Chocolate"	"Chocolate"
##	[77,]	"Cookie Dough"	"Chocolate"	"Chocolate"
##	[78,]	"Mint Chocolate Chip"	"Chocolate"	"Chocolate"
##	[79,]	"Butter Pecan"	"Chocolate"	"Chocolate"
##	[80,]	"Coffee"	"Chocolate"	"Chocolate"
##	[81,]	"Vanilla"	"Strawberry"	"Chocolate"
##	[82,]	"Chocolate"	"Strawberry"	"Chocolate"
##	[83,]	"Strawberry"	"Strawberry"	"Chocolate"
##	[84,]	"Cookies & Cream"	"Strawberry"	"Chocolate"
##	[85,]	"Cookie Dough"	"Strawberry"	"Chocolate"
##	[86,]	"Mint Chocolate Chip"	"Strawberry"	"Chocolate"
##	[87,]	"Butter Pecan"	"Strawberry"	"Chocolate"
##	[88,]	"Coffee"	"Strawberry"	"Chocolate"
##	[89,]	"Vanilla"	"Cookies & Cream"	"Chocolate"
##	[90,]	"Chocolate"	"Cookies & Cream"	"Chocolate"
##	[91,]	"Strawberry"	"Cookies & Cream"	"Chocolate"
##	[92,]	"Cookies & Cream"	"Cookies & Cream"	"Chocolate"

##	[93,]	"Cookie Dough"	"Cookies & Cream"	"Chocolate"
##	[94,]	"Mint Chocolate Chip"	"Cookies & Cream"	"Chocolate"
##	[95,]	"Butter Pecan"	"Cookies & Cream"	"Chocolate"
##	[96,]	"Coffee"	"Cookies & Cream"	"Chocolate"
##	[97,]	"Vanilla"	"Cookie Dough"	"Chocolate"
##	[98,]	"Chocolate"	"Cookie Dough"	"Chocolate"
##	[99,]	"Strawberry"	"Cookie Dough"	"Chocolate"
##	[100,]	"Cookies & Cream"	"Cookie Dough"	"Chocolate"
##	[101,]	"Cookie Dough"	"Cookie Dough"	"Chocolate"
##	[102,]	"Mint Chocolate Chip"	"Cookie Dough"	"Chocolate"
##	[103,]	"Butter Pecan"	"Cookie Dough"	"Chocolate"
##	[104,]	"Coffee"	"Cookie Dough"	"Chocolate"
##	[105,]	"Vanilla"	"Mint Chocolate Chip"	"Chocolate"
##	[106,]	"Chocolate"	"Mint Chocolate Chip"	"Chocolate"
##	[107,]	"Strawberry"	"Mint Chocolate Chip"	"Chocolate"
##	[108,]	"Cookies & Cream"	"Mint Chocolate Chip"	"Chocolate"
##	[109,]	"Cookie Dough"	"Mint Chocolate Chip"	"Chocolate"
##	[110,]	"Mint Chocolate Chip"	"Mint Chocolate Chip"	"Chocolate"
##	[111,]	"Butter Pecan"	"Mint Chocolate Chip"	"Chocolate"
##	[112,]	"Coffee"	"Mint Chocolate Chip"	"Chocolate"
##	[113,]	"Vanilla"	"Butter Pecan"	"Chocolate"
##	[114,]	"Chocolate"	"Butter Pecan"	"Chocolate"
##	[115,]	"Strawberry"	"Butter Pecan"	"Chocolate"
##	[116,]	"Cookies & Cream"	"Butter Pecan"	"Chocolate"
##	[117,]	"Cookie Dough"	"Butter Pecan"	"Chocolate"
##	[118,]	"Mint Chocolate Chip"	"Butter Pecan"	"Chocolate"
##	[119,]	"Butter Pecan"	"Butter Pecan"	"Chocolate"
##	[120,]	"Coffee"	"Butter Pecan"	"Chocolate"
##	[121,]	"Vanilla"	"Coffee"	"Chocolate"
##	[122,]	"Chocolate"	"Coffee"	"Chocolate"
##	[123,]	"Strawberry"	"Coffee"	"Chocolate"
##	[124,]	"Cookies & Cream"	"Coffee"	"Chocolate"
##	[125,]	"Cookie Dough"	"Coffee"	"Chocolate"
##	[126,]	"Mint Chocolate Chip"	"Coffee"	"Chocolate"
##	[127,]	"Butter Pecan"	"Coffee"	"Chocolate"
##	[128,]	"Coffee"	"Coffee"	"Chocolate"
##	[129,]	"Vanilla"	"Vanilla"	"Strawberry"
##	[130,]	"Chocolate"	"Vanilla"	"Strawberry"
##	[131,]	"Strawberry"	"Vanilla"	"Strawberry"
##	[132,]	"Cookies & Cream"	"Vanilla"	"Strawberry"
##	[133,]	"Cookie Dough"	"Vanilla"	"Strawberry"
##	[134,]	"Mint Chocolate Chip"	"Vanilla"	"Strawberry"
##	[135,]	"Butter Pecan"	"Vanilla"	"Strawberry"
##	[136,]	"Coffee"	"Vanilla"	"Strawberry"
##	[137,]	"Vanilla"	"Chocolate"	"Strawberry"
##	[138,]	"Chocolate"	"Chocolate"	"Strawberry"
##	[139,]	"Strawberry"	"Chocolate"	"Strawberry"
##	[140,]	"Cookies & Cream"	"Chocolate"	"Strawberry"
##	[141,]	"Cookie Dough"	"Chocolate"	"Strawberry"
##	[142,]	"Mint Chocolate Chip"	"Chocolate"	"Strawberry"
##	[143,]	"Butter Pecan"	"Chocolate"	"Strawberry"
##	[144,]	"Coffee"	"Chocolate"	"Strawberry"
##	[145,]	"Vanilla"	"Strawberry"	"Strawberry"
##	[146,]	"Chocolate"	"Strawberry"	"Strawberry"

## [147,]	"Strawberry"	"Strawberry"	"Strawberry"
## [148,]	"Cookies & Cream"	"Strawberry"	"Strawberry"
## [149,]	"Cookie Dough"	"Strawberry"	"Strawberry"
## [150,]	"Mint Chocolate Chip"	"Strawberry"	"Strawberry"
## [151,]	"Butter Pecan"	"Strawberry"	"Strawberry"
## [152,]	"Coffee"	"Strawberry"	"Strawberry"
## [153,]	"Vanilla"	"Cookies & Cream"	"Strawberry"
## [154,]	"Chocolate"	"Cookies & Cream"	"Strawberry"
## [155,]	"Strawberry"	"Cookies & Cream"	"Strawberry"
## [156,]	"Cookies & Cream"	"Cookies & Cream"	"Strawberry"
## [157,]	"Cookie Dough"	"Cookies & Cream"	"Strawberry"
## [158,]	"Mint Chocolate Chip"	"Cookies & Cream"	"Strawberry"
## [159,]	"Butter Pecan"	"Cookies & Cream"	"Strawberry"
## [160,]	"Coffee"	"Cookies & Cream"	"Strawberry"
## [161,]	"Vanilla"	"Cookie Dough"	"Strawberry"
## [162,]	"Chocolate"	"Cookie Dough"	"Strawberry"
## [163,]	"Strawberry"	"Cookie Dough"	"Strawberry"
## [164,]	"Cookies & Cream"	"Cookie Dough"	"Strawberry"
## [165,]	"Cookie Dough"	"Cookie Dough"	"Strawberry"
## [166,]	"Mint Chocolate Chip"	"Cookie Dough"	"Strawberry"
## [167,]	"Butter Pecan"	"Cookie Dough"	"Strawberry"
## [168,]	"Coffee"	"Cookie Dough"	"Strawberry"
## [169,]	"Vanilla"	"Mint Chocolate Chip"	"Strawberry"
## [170,]	"Chocolate"	"Mint Chocolate Chip"	"Strawberry"
## [171,]	"Strawberry"	"Mint Chocolate Chip"	"Strawberry"
## [172,]	"Cookies & Cream"	"Mint Chocolate Chip"	"Strawberry"
## [173,]	"Cookie Dough"	"Mint Chocolate Chip"	"Strawberry"
## [174,]	"Mint Chocolate Chip"	"Mint Chocolate Chip"	"Strawberry"
## [175,]	"Butter Pecan"	"Mint Chocolate Chip"	"Strawberry"
## [176,]	"Coffee"	"Mint Chocolate Chip"	"Strawberry"
## [177,]	"Vanilla"	"Butter Pecan"	"Strawberry"
## [178,]	"Chocolate"	"Butter Pecan"	"Strawberry"
## [179,]	"Strawberry"	"Butter Pecan"	"Strawberry"
## [180,]	"Cookies & Cream"	"Butter Pecan"	"Strawberry"
## [181,]	"Cookie Dough"	"Butter Pecan"	"Strawberry"
## [182,]	"Mint Chocolate Chip"	"Butter Pecan"	"Strawberry"
## [183,]	"Butter Pecan"	"Butter Pecan"	"Strawberry"
## [184,]	"Coffee"	"Butter Pecan"	"Strawberry"
## [185,]	"Vanilla"	"Coffee"	"Strawberry"
## [186,]	"Chocolate"	"Coffee"	"Strawberry"
## [187,]	"Strawberry"	"Coffee"	"Strawberry"
## [188,]	"Cookies & Cream"	"Coffee"	"Strawberry"
## [189,]	"Cookie Dough"	"Coffee"	"Strawberry"
## [190,]	"Mint Chocolate Chip"	"Coffee"	"Strawberry"
## [191,]	"Butter Pecan"	"Coffee"	"Strawberry"
## [192,]	"Coffee"	"Coffee"	"Strawberry"
## [193,]	"Vanilla"	"Vanilla"	"Cookies & Cream"
## [194,]	"Chocolate"	"Vanilla"	"Cookies & Cream"
## [195,]	"Strawberry"	"Vanilla"	"Cookies & Cream"
## [196,]	"Cookies & Cream"	"Vanilla"	"Cookies & Cream"
## [197,]	"Cookie Dough"	"Vanilla"	"Cookies & Cream"
## [198,]	"Mint Chocolate Chip"	"Vanilla"	"Cookies & Cream"
## [199,]	"Butter Pecan"	"Vanilla"	"Cookies & Cream"
## [200,]	"Coffee"	"Vanilla"	"Cookies & Cream"

## [201,]	"Vanilla"	"Chocolate"	"Cookies & Cream"
## [202,]	"Chocolate"	"Chocolate"	"Cookies & Cream"
## [203,]	"Strawberry"	"Chocolate"	"Cookies & Cream"
## [204,]	"Cookies & Cream"	"Chocolate"	"Cookies & Cream"
## [205,]	"Cookie Dough"	"Chocolate"	"Cookies & Cream"
## [206,]	"Mint Chocolate Chip"	"Chocolate"	"Cookies & Cream"
## [207,]	"Butter Pecan"	"Chocolate"	"Cookies & Cream"
## [208,]	"Coffee"	"Chocolate"	"Cookies & Cream"
## [209,]	"Vanilla"	"Strawberry"	"Cookies & Cream"
## [210,]	"Chocolate"	"Strawberry"	"Cookies & Cream"
## [211,]	"Strawberry"	"Strawberry"	"Cookies & Cream"
## [212,]	"Cookies & Cream"	"Strawberry"	"Cookies & Cream"
## [213,]	"Cookie Dough"	"Strawberry"	"Cookies & Cream"
## [214,]	"Mint Chocolate Chip"	"Strawberry"	"Cookies & Cream"
## [215,]	"Butter Pecan"	"Strawberry"	"Cookies & Cream"
## [216,]	"Coffee"	"Strawberry"	"Cookies & Cream"
## [217,]	"Vanilla"	"Cookies & Cream"	"Cookies & Cream"
## [218,]	"Chocolate"	"Cookies & Cream"	"Cookies & Cream"
## [219,]	"Strawberry"	"Cookies & Cream"	"Cookies & Cream"
## [220,]	"Cookies & Cream"	"Cookies & Cream"	"Cookies & Cream"
## [221,]	"Cookie Dough"	"Cookies & Cream"	"Cookies & Cream"
## [222,]	"Mint Chocolate Chip"	"Cookies & Cream"	"Cookies & Cream"
## [223,]	"Butter Pecan"	"Cookies & Cream"	"Cookies & Cream"
## [224,]	"Coffee"	"Cookies & Cream"	"Cookies & Cream"
## [225,]	"Vanilla"	"Cookie Dough"	"Cookies & Cream"
## [226,]	"Chocolate"	"Cookie Dough"	"Cookies & Cream"
## [227,]	"Strawberry"	"Cookie Dough"	"Cookies & Cream"
## [228,]	"Cookies & Cream"	"Cookie Dough"	"Cookies & Cream"
## [229,]	"Cookie Dough"	"Cookie Dough"	"Cookies & Cream"
## [230,]	"Mint Chocolate Chip"	"Cookie Dough"	"Cookies & Cream"
## [231,]	"Butter Pecan"	"Cookie Dough"	"Cookies & Cream"
## [232,]	"Coffee"	"Cookie Dough"	"Cookies & Cream"
## [233,]	"Vanilla"	"Mint Chocolate Chip"	"Cookies & Cream"
## [234,]	"Chocolate"	"Mint Chocolate Chip"	"Cookies & Cream"
## [235,]	"Strawberry"	"Mint Chocolate Chip"	"Cookies & Cream"
## [236,]	"Cookies & Cream"	"Mint Chocolate Chip"	"Cookies & Cream"
## [237,]	"Cookie Dough"	"Mint Chocolate Chip"	"Cookies & Cream"
## [238,]	"Mint Chocolate Chip"	"Mint Chocolate Chip"	"Cookies & Cream"
## [239,]	"Butter Pecan"	"Mint Chocolate Chip"	"Cookies & Cream"
## [240,]	"Coffee"	"Mint Chocolate Chip"	"Cookies & Cream"
## [241,]	"Vanilla"	"Butter Pecan"	"Cookies & Cream"
## [242,]	"Chocolate"	"Butter Pecan"	"Cookies & Cream"
## [243,]	"Strawberry"	"Butter Pecan"	"Cookies & Cream"
## [244,]	"Cookies & Cream"	"Butter Pecan"	"Cookies & Cream"
## [245,]	"Cookie Dough"	"Butter Pecan"	"Cookies & Cream"
## [246,]	"Mint Chocolate Chip"	"Butter Pecan"	"Cookies & Cream"
## [247,]	"Butter Pecan"	"Butter Pecan"	"Cookies & Cream"
## [248,]	"Coffee"	"Butter Pecan"	"Cookies & Cream"
## [249,]	"Vanilla"	"Coffee"	"Cookies & Cream"
## [250,]	"Chocolate"	"Coffee"	"Cookies & Cream"
## [251,]	"Strawberry"	"Coffee"	"Cookies & Cream"
## [252,]	"Cookies & Cream"	"Coffee"	"Cookies & Cream"
## [253,]	"Cookie Dough"	"Coffee"	"Cookies & Cream"
## [254,]	"Mint Chocolate Chip"	"Coffee"	"Cookies & Cream"

## [255,]	"Butter Pecan"	"Coffee"	"Cookies & Cream"
## [256,]	"Coffee"	"Coffee"	"Cookies & Cream"
## [257,]	"Vanilla"	"Vanilla"	"Cookie Dough"
## [258,]	"Chocolate"	"Vanilla"	"Cookie Dough"
## [259,]	"Strawberry"	"Vanilla"	"Cookie Dough"
## [260,]	"Cookies & Cream"	"Vanilla"	"Cookie Dough"
## [261,]	"Cookie Dough"	"Vanilla"	"Cookie Dough"
## [262,]	"Mint Chocolate Chip"	"Vanilla"	"Cookie Dough"
## [263,]	"Butter Pecan"	"Vanilla"	"Cookie Dough"
## [264,]	"Coffee"	"Vanilla"	"Cookie Dough"
## [265,]	"Vanilla"	"Chocolate"	"Cookie Dough"
## [266,]	"Chocolate"	"Chocolate"	"Cookie Dough"
## [267,]	"Strawberry"	"Chocolate"	"Cookie Dough"
## [268,]	"Cookies & Cream"	"Chocolate"	"Cookie Dough"
## [269,]	"Cookie Dough"	"Chocolate"	"Cookie Dough"
## [270,]	"Mint Chocolate Chip"	"Chocolate"	"Cookie Dough"
## [271,]	"Butter Pecan"	"Chocolate"	"Cookie Dough"
## [272,]	"Coffee"	"Chocolate"	"Cookie Dough"
## [273,]	"Vanilla"	"Strawberry"	"Cookie Dough"
## [274,]	"Chocolate"	"Strawberry"	"Cookie Dough"
## [275,]	"Strawberry"	"Strawberry"	"Cookie Dough"
## [276,]	"Cookies & Cream"	"Strawberry"	"Cookie Dough"
## [277,]	"Cookie Dough"	"Strawberry"	"Cookie Dough"
## [278,]	"Mint Chocolate Chip"	"Strawberry"	"Cookie Dough"
## [279,]	"Butter Pecan"	"Strawberry"	"Cookie Dough"
## [280,]	"Coffee"	"Strawberry"	"Cookie Dough"
## [281,]	"Vanilla"	"Cookies & Cream"	"Cookie Dough"
## [282,]	"Chocolate"	"Cookies & Cream"	"Cookie Dough"
## [283,]	"Strawberry"	"Cookies & Cream"	"Cookie Dough"
## [284,]	"Cookies & Cream"	"Cookies & Cream"	"Cookie Dough"
## [285,]	"Cookie Dough"	"Cookies & Cream"	"Cookie Dough"
## [286,]	"Mint Chocolate Chip"	"Cookies & Cream"	"Cookie Dough"
## [287,]	"Butter Pecan"	"Cookies & Cream"	"Cookie Dough"
## [288,]	"Coffee"	"Cookies & Cream"	"Cookie Dough"
## [289,]	"Vanilla"	"Cookie Dough"	"Cookie Dough"
## [290,]	"Chocolate"	"Cookie Dough"	"Cookie Dough"
## [291,]	"Strawberry"	"Cookie Dough"	"Cookie Dough"
## [292,]	"Cookies & Cream"	"Cookie Dough"	"Cookie Dough"
## [293,]	"Cookie Dough"	"Cookie Dough"	"Cookie Dough"
## [294,]	"Mint Chocolate Chip"	"Cookie Dough"	"Cookie Dough"
## [295,]	"Butter Pecan"	"Cookie Dough"	"Cookie Dough"
## [296,]	"Coffee"	"Cookie Dough"	"Cookie Dough"
## [297,]	"Vanilla"	"Mint Chocolate Chip"	"Cookie Dough"
## [298,]	"Chocolate"	"Mint Chocolate Chip"	"Cookie Dough"
## [299,]	"Strawberry"	"Mint Chocolate Chip"	"Cookie Dough"
## [300,]	"Cookies & Cream"	"Mint Chocolate Chip"	"Cookie Dough"
## [301,]	"Cookie Dough"	"Mint Chocolate Chip"	"Cookie Dough"
## [302,]	"Mint Chocolate Chip"	"Mint Chocolate Chip"	"Cookie Dough"
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## [304,]	"Coffee"	"Mint Chocolate Chip"	"Cookie Dough"
## [305,]	"Vanilla"	"Butter Pecan"	"Cookie Dough"
## [306,]	"Chocolate"	"Butter Pecan"	"Cookie Dough"
## [307,]	"Strawberry"	"Butter Pecan"	"Cookie Dough"
## [308,]	"Cookies & Cream"	"Butter Pecan"	"Cookie Dough"

## [309,]	"Cookie Dough"	"Butter Pecan"	"Cookie Dough"
## [310,]	"Mint Chocolate Chip"	"Butter Pecan"	"Cookie Dough"
## [311,]	"Butter Pecan"	"Butter Pecan"	"Cookie Dough"
## [312,]	"Coffee"	"Butter Pecan"	"Cookie Dough"
## [313,]	"Vanilla"	"Coffee"	"Cookie Dough"
## [314,]	"Chocolate"	"Coffee"	"Cookie Dough"
## [315,]	"Strawberry"	"Coffee"	"Cookie Dough"
## [316,]	"Cookies & Cream"	"Coffee"	"Cookie Dough"
## [317,]	"Cookie Dough"	"Coffee"	"Cookie Dough"
## [318,]	"Mint Chocolate Chip"	"Coffee"	"Cookie Dough"
## [319,]	"Butter Pecan"	"Coffee"	"Cookie Dough"
## [320,]	"Coffee"	"Coffee"	"Cookie Dough"
## [321,]	"Vanilla"	"Vanilla"	"Mint Chocolate Chip"
## [322,]	"Chocolate"	"Vanilla"	"Mint Chocolate Chip"
## [323,]	"Strawberry"	"Vanilla"	"Mint Chocolate Chip"
## [324,]	"Cookies & Cream"	"Vanilla"	"Mint Chocolate Chip"
## [325,]	"Cookie Dough"	"Vanilla"	"Mint Chocolate Chip"
## [326,]	"Mint Chocolate Chip"	"Vanilla"	"Mint Chocolate Chip"
## [327,]	"Butter Pecan"	"Vanilla"	"Mint Chocolate Chip"
## [328,]	"Coffee"	"Vanilla"	"Mint Chocolate Chip"
## [329,]	"Vanilla"	"Chocolate"	"Mint Chocolate Chip"
## [330,]	"Chocolate"	"Chocolate"	"Mint Chocolate Chip"
## [331,]	"Strawberry"	"Chocolate"	"Mint Chocolate Chip"
## [332,]	"Cookies & Cream"	"Chocolate"	"Mint Chocolate Chip"
## [333,]	"Cookie Dough"	"Chocolate"	"Mint Chocolate Chip"
## [334,]	"Mint Chocolate Chip"	"Chocolate"	"Mint Chocolate Chip"
## [335,]	"Butter Pecan"	"Chocolate"	"Mint Chocolate Chip"
## [336,]	"Coffee"	"Chocolate"	"Mint Chocolate Chip"
## [337,]	"Vanilla"	"Strawberry"	"Mint Chocolate Chip"
## [338,]	"Chocolate"	"Strawberry"	"Mint Chocolate Chip"
## [339,]	"Strawberry"	"Strawberry"	"Mint Chocolate Chip"
## [340,]	"Cookies & Cream"	"Strawberry"	"Mint Chocolate Chip"
## [341,]	"Cookie Dough"	"Strawberry"	"Mint Chocolate Chip"
## [342,]	"Mint Chocolate Chip"	"Strawberry"	"Mint Chocolate Chip"
## [343,]	"Butter Pecan"	"Strawberry"	"Mint Chocolate Chip"
## [344,]	"Coffee"	"Strawberry"	"Mint Chocolate Chip"
## [345,]	"Vanilla"	"Cookies & Cream"	"Mint Chocolate Chip"
## [346,]	"Chocolate"	"Cookies & Cream"	"Mint Chocolate Chip"
## [347,]	"Strawberry"	"Cookies & Cream"	"Mint Chocolate Chip"
## [348,]	"Cookies & Cream"	"Cookies & Cream"	"Mint Chocolate Chip"
## [349,]	"Cookie Dough"	"Cookies & Cream"	"Mint Chocolate Chip"
## [350,]	"Mint Chocolate Chip"	"Cookies & Cream"	"Mint Chocolate Chip"
## [351,]	"Butter Pecan"	"Cookies & Cream"	"Mint Chocolate Chip"
## [352,]	"Coffee"	"Cookies & Cream"	"Mint Chocolate Chip"
## [353,]	"Vanilla"	"Cookie Dough"	"Mint Chocolate Chip"
## [354,]	"Chocolate"	"Cookie Dough"	"Mint Chocolate Chip"
## [355,]	"Strawberry"	"Cookie Dough"	"Mint Chocolate Chip"
## [356,]	"Cookies & Cream"	"Cookie Dough"	"Mint Chocolate Chip"
## [357,]	"Cookie Dough"	"Cookie Dough"	"Mint Chocolate Chip"
## [358,]	"Mint Chocolate Chip"	"Cookie Dough"	"Mint Chocolate Chip"
## [359,]	"Butter Pecan"	"Cookie Dough"	"Mint Chocolate Chip"
## [360,]	"Coffee"	"Cookie Dough"	"Mint Chocolate Chip"
## [361,]	"Vanilla"	"Mint Chocolate Chip"	"Mint Chocolate Chip"
## [362,]	"Chocolate"	"Mint Chocolate Chip"	"Mint Chocolate Chip"

## [363,]	"Strawberry"	"Mint Chocolate Chip"	"Mint Chocolate Chip"
## [364,]	"Cookies & Cream"	"Mint Chocolate Chip"	"Mint Chocolate Chip"
## [365,]	"Cookie Dough"	"Mint Chocolate Chip"	"Mint Chocolate Chip"
## [366,]	"Mint Chocolate Chip"	"Mint Chocolate Chip"	"Mint Chocolate Chip"
## [367,]	"Butter Pecan"	"Mint Chocolate Chip"	"Mint Chocolate Chip"
## [368,]	"Coffee"	"Mint Chocolate Chip"	"Mint Chocolate Chip"
## [369,]	"Vanilla"	"Butter Pecan"	"Mint Chocolate Chip"
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## [373,]	"Cookie Dough"	"Butter Pecan"	"Mint Chocolate Chip"
## [374,]	"Mint Chocolate Chip"	"Butter Pecan"	"Mint Chocolate Chip"
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## [376,]	"Coffee"	"Butter Pecan"	"Mint Chocolate Chip"
## [377,]	"Vanilla"	"Coffee"	"Mint Chocolate Chip"
## [378,]	"Chocolate"	"Coffee"	"Mint Chocolate Chip"
## [379,]	"Strawberry"	"Coffee"	"Mint Chocolate Chip"
## [380,]	"Cookies & Cream"	"Coffee"	"Mint Chocolate Chip"
## [381,]	"Cookie Dough"	"Coffee"	"Mint Chocolate Chip"
## [382,]	"Mint Chocolate Chip"	"Coffee"	"Mint Chocolate Chip"
## [383,]	"Butter Pecan"	"Coffee"	"Mint Chocolate Chip"
## [384,]	"Coffee"	"Coffee"	"Mint Chocolate Chip"
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## [388,]	"Cookies & Cream"	"Vanilla"	"Butter Pecan"
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## [390,]	"Mint Chocolate Chip"	"Vanilla"	"Butter Pecan"
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## [393,]	"Vanilla"	"Chocolate"	"Butter Pecan"
## [394,]	"Chocolate"	"Chocolate"	"Butter Pecan"
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## [396,]	"Cookies & Cream"	"Chocolate"	"Butter Pecan"
## [397,]	"Cookie Dough"	"Chocolate"	"Butter Pecan"
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## [409,]	"Vanilla"	"Cookies & Cream"	"Butter Pecan"
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## [411,]	"Strawberry"	"Cookies & Cream"	"Butter Pecan"
## [412,]	"Cookies & Cream"	"Cookies & Cream"	"Butter Pecan"
## [413,]	"Cookie Dough"	"Cookies & Cream"	"Butter Pecan"
## [414,]	"Mint Chocolate Chip"	"Cookies & Cream"	"Butter Pecan"
## [415,]	"Butter Pecan"	"Cookies & Cream"	"Butter Pecan"
## [416,]	"Coffee"	"Cookies & Cream"	"Butter Pecan"

## [417,]	"Vanilla"	"Cookie Dough"	"Butter Pecan"
## [418,]	"Chocolate"	"Cookie Dough"	"Butter Pecan"
## [419,]	"Strawberry"	"Cookie Dough"	"Butter Pecan"
## [420,]	"Cookies & Cream"	"Cookie Dough"	"Butter Pecan"
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## [422,]	"Mint Chocolate Chip"	"Cookie Dough"	"Butter Pecan"
## [423,]	"Butter Pecan"	"Cookie Dough"	"Butter Pecan"
## [424,]	"Coffee"	"Cookie Dough"	"Butter Pecan"
## [425,]	"Vanilla"	"Mint Chocolate Chip"	"Butter Pecan"
## [426,]	"Chocolate"	"Mint Chocolate Chip"	"Butter Pecan"
## [427,]	"Strawberry"	"Mint Chocolate Chip"	"Butter Pecan"
## [428,]	"Cookies & Cream"	"Mint Chocolate Chip"	"Butter Pecan"
## [429,]	"Cookie Dough"	"Mint Chocolate Chip"	"Butter Pecan"
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## [433,]	"Vanilla"	"Butter Pecan"	"Butter Pecan"
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## [435,]	"Strawberry"	"Butter Pecan"	"Butter Pecan"
## [436,]	"Cookies & Cream"	"Butter Pecan"	"Butter Pecan"
## [437,]	"Cookie Dough"	"Butter Pecan"	"Butter Pecan"
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## [441,]	"Vanilla"	"Coffee"	"Butter Pecan"
## [442,]	"Chocolate"	"Coffee"	"Butter Pecan"
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## [450,]	"Chocolate"	"Vanilla"	"Coffee"
## [451,]	"Strawberry"	"Vanilla"	"Coffee"
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## [456,]	"Coffee"	"Vanilla"	"Coffee"
## [457,]	"Vanilla"	"Chocolate"	"Coffee"
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## [461,]	"Cookie Dough"	"Chocolate"	"Coffee"
## [462,]	"Mint Chocolate Chip"	"Chocolate"	"Coffee"
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## [464,]	"Coffee"	"Chocolate"	"Coffee"
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## [466,]	"Chocolate"	"Strawberry"	"Coffee"
## [467,]	"Strawberry"	"Strawberry"	"Coffee"
## [468,]	"Cookies & Cream"	"Strawberry"	"Coffee"
## [469,]	"Cookie Dough"	"Strawberry"	"Coffee"
## [470,]	"Mint Chocolate Chip"	"Strawberry"	"Coffee"

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## [472,] "Coffee"           "Strawberry"      "Coffee"
## [473,] "Vanilla"          "Cookies & Cream" "Coffee"
## [474,] "Chocolate"        "Cookies & Cream" "Coffee"
## [475,] "Strawberry"       "Cookies & Cream" "Coffee"
## [476,] "Cookies & Cream"   "Cookies & Cream" "Coffee"
## [477,] "Cookie Dough"     "Cookies & Cream" "Coffee"
## [478,] "Mint Chocolate Chip" "Cookies & Cream" "Coffee"
## [479,] "Butter Pecan"     "Cookies & Cream" "Coffee"
## [480,] "Coffee"           "Cookies & Cream" "Coffee"
## [481,] "Vanilla"          "Cookie Dough"    "Coffee"
## [482,] "Chocolate"        "Cookie Dough"    "Coffee"
## [483,] "Strawberry"       "Cookie Dough"    "Coffee"
## [484,] "Cookies & Cream"   "Cookie Dough"    "Coffee"
## [485,] "Cookie Dough"     "Cookie Dough"    "Coffee"
## [486,] "Mint Chocolate Chip" "Cookie Dough"    "Coffee"
## [487,] "Butter Pecan"     "Cookie Dough"    "Coffee"
## [488,] "Coffee"           "Cookie Dough"    "Coffee"
## [489,] "Vanilla"          "Mint Chocolate Chip" "Coffee"
## [490,] "Chocolate"        "Mint Chocolate Chip" "Coffee"
## [491,] "Strawberry"       "Mint Chocolate Chip" "Coffee"
## [492,] "Cookies & Cream"   "Mint Chocolate Chip" "Coffee"
## [493,] "Cookie Dough"     "Mint Chocolate Chip" "Coffee"
## [494,] "Mint Chocolate Chip" "Mint Chocolate Chip" "Coffee"
## [495,] "Butter Pecan"     "Mint Chocolate Chip" "Coffee"
## [496,] "Coffee"           "Mint Chocolate Chip" "Coffee"
## [497,] "Vanilla"          "Butter Pecan"    "Coffee"
## [498,] "Chocolate"        "Butter Pecan"    "Coffee"
## [499,] "Strawberry"       "Butter Pecan"    "Coffee"
## [500,] "Cookies & Cream"   "Butter Pecan"    "Coffee"
## [501,] "Cookie Dough"     "Butter Pecan"    "Coffee"
## [502,] "Mint Chocolate Chip" "Butter Pecan"    "Coffee"
## [503,] "Butter Pecan"     "Butter Pecan"    "Coffee"
## [504,] "Coffee"           "Butter Pecan"    "Coffee"
## [505,] "Vanilla"          "Coffee"           "Coffee"
## [506,] "Chocolate"        "Coffee"           "Coffee"
## [507,] "Strawberry"       "Coffee"           "Coffee"
## [508,] "Cookies & Cream"   "Coffee"           "Coffee"
## [509,] "Cookie Dough"     "Coffee"           "Coffee"
## [510,] "Mint Chocolate Chip" "Coffee"           "Coffee"
## [511,] "Butter Pecan"     "Coffee"           "Coffee"
## [512,] "Coffee"           "Coffee"           "Coffee"
```

```
print(head(Scenario3, 10)) # First 10
```

```
##      [,1]      [,2]      [,3]
## [1,] "Vanilla"  "Vanilla"  "Vanilla"
## [2,] "Chocolate" "Vanilla"  "Vanilla"
## [3,] "Strawberry" "Vanilla"  "Vanilla"
## [4,] "Cookies & Cream" "Vanilla"  "Vanilla"
## [5,] "Cookie Dough" "Vanilla"  "Vanilla"
## [6,] "Mint Chocolate Chip" "Vanilla"  "Vanilla"
## [7,] "Butter Pecan" "Vanilla"  "Vanilla"
## [8,] "Coffee"    "Vanilla"  "Vanilla"
## [9,] "Vanilla"   "Chocolate" "Vanilla"
```

```
## [10,] "Chocolate"          "Chocolate" "Vanilla"
print(tail(Scenario3, 10)) # Last 10

##          [,1]          [,2]          [,3]
## [503,] "Butter Pecan"    "Butter Pecan" "Coffee"
## [504,] "Coffee"         "Butter Pecan" "Coffee"
## [505,] "Vanilla"        "Coffee"       "Coffee"
## [506,] "Chocolate"      "Coffee"       "Coffee"
## [507,] "Strawberry"     "Coffee"       "Coffee"
## [508,] "Cookies & Cream" "Coffee"       "Coffee"
## [509,] "Cookie Dough"   "Coffee"       "Coffee"
## [510,] "Mint Chocolate Chip" "Coffee"     "Coffee"
## [511,] "Butter Pecan"   "Coffee"       "Coffee"
## [512,] "Coffee"         "Coffee"       "Coffee"

print(Scenario3[sample(nrow(Scenario3), 5), ]) # 5 Random

##          [,1]          [,2]          [,3]
## [1,] "Cookie Dough"    "Mint Chocolate Chip" "Butter Pecan"
## [2,] "Coffee"          "Cookies & Cream"    "Butter Pecan"
## [3,] "Vanilla"         "Vanilla"           "Strawberry"
## [4,] "Cookies & Cream" "Vanilla"           "Cookies & Cream"
## [5,] "Chocolate"       "Mint Chocolate Chip" "Coffee"

#512 total combos in first print
print(8^3) #8^3=512

## [1] 512

#Same
```

Scenario 4: Unordered, with replacement

```
Scenario4 <- CombSet(Flavors, 3, repl = TRUE, ord = FALSE)
print(Scenario4) # Print Total

##          [,1]          [,2]          [,3]
## [1,] "Vanilla"        "Vanilla"        "Vanilla"
## [2,] "Chocolate"      "Vanilla"        "Vanilla"
## [3,] "Strawberry"     "Vanilla"        "Vanilla"
## [4,] "Cookies & Cream" "Vanilla"        "Vanilla"
## [5,] "Cookie Dough"   "Vanilla"        "Vanilla"
## [6,] "Mint Chocolate Chip" "Vanilla"      "Vanilla"
## [7,] "Butter Pecan"   "Vanilla"        "Vanilla"
## [8,] "Coffee"         "Vanilla"        "Vanilla"
## [9,] "Chocolate"      "Chocolate"      "Vanilla"
## [10,] "Chocolate"     "Strawberry"     "Vanilla"
## [11,] "Chocolate"     "Cookies & Cream" "Vanilla"
## [12,] "Chocolate"     "Cookie Dough"   "Vanilla"
## [13,] "Chocolate"     "Mint Chocolate Chip" "Vanilla"
## [14,] "Butter Pecan"   "Chocolate"      "Vanilla"
## [15,] "Chocolate"     "Coffee"         "Vanilla"
## [16,] "Strawberry"    "Strawberry"     "Vanilla"
## [17,] "Cookies & Cream" "Strawberry"     "Vanilla"
## [18,] "Cookie Dough"   "Strawberry"     "Vanilla"
```

##	[19,]	"Mint Chocolate Chip"	"Strawberry"	"Vanilla"
##	[20,]	"Butter Pecan"	"Strawberry"	"Vanilla"
##	[21,]	"Coffee"	"Strawberry"	"Vanilla"
##	[22,]	"Cookies & Cream"	"Cookies & Cream"	"Vanilla"
##	[23,]	"Cookie Dough"	"Cookies & Cream"	"Vanilla"
##	[24,]	"Cookies & Cream"	"Mint Chocolate Chip"	"Vanilla"
##	[25,]	"Butter Pecan"	"Cookies & Cream"	"Vanilla"
##	[26,]	"Coffee"	"Cookies & Cream"	"Vanilla"
##	[27,]	"Cookie Dough"	"Cookie Dough"	"Vanilla"
##	[28,]	"Cookie Dough"	"Mint Chocolate Chip"	"Vanilla"
##	[29,]	"Butter Pecan"	"Cookie Dough"	"Vanilla"
##	[30,]	"Coffee"	"Cookie Dough"	"Vanilla"
##	[31,]	"Mint Chocolate Chip"	"Mint Chocolate Chip"	"Vanilla"
##	[32,]	"Butter Pecan"	"Mint Chocolate Chip"	"Vanilla"
##	[33,]	"Coffee"	"Mint Chocolate Chip"	"Vanilla"
##	[34,]	"Butter Pecan"	"Butter Pecan"	"Vanilla"
##	[35,]	"Butter Pecan"	"Coffee"	"Vanilla"
##	[36,]	"Coffee"	"Coffee"	"Vanilla"
##	[37,]	"Chocolate"	"Chocolate"	"Chocolate"
##	[38,]	"Chocolate"	"Chocolate"	"Strawberry"
##	[39,]	"Chocolate"	"Chocolate"	"Cookies & Cream"
##	[40,]	"Chocolate"	"Chocolate"	"Cookie Dough"
##	[41,]	"Chocolate"	"Chocolate"	"Mint Chocolate Chip"
##	[42,]	"Butter Pecan"	"Chocolate"	"Chocolate"
##	[43,]	"Chocolate"	"Chocolate"	"Coffee"
##	[44,]	"Chocolate"	"Strawberry"	"Strawberry"
##	[45,]	"Chocolate"	"Cookies & Cream"	"Strawberry"
##	[46,]	"Chocolate"	"Cookie Dough"	"Strawberry"
##	[47,]	"Chocolate"	"Mint Chocolate Chip"	"Strawberry"
##	[48,]	"Butter Pecan"	"Chocolate"	"Strawberry"
##	[49,]	"Chocolate"	"Coffee"	"Strawberry"
##	[50,]	"Chocolate"	"Cookies & Cream"	"Cookies & Cream"
##	[51,]	"Chocolate"	"Cookie Dough"	"Cookies & Cream"
##	[52,]	"Chocolate"	"Cookies & Cream"	"Mint Chocolate Chip"
##	[53,]	"Butter Pecan"	"Chocolate"	"Cookies & Cream"
##	[54,]	"Chocolate"	"Coffee"	"Cookies & Cream"
##	[55,]	"Chocolate"	"Cookie Dough"	"Cookie Dough"
##	[56,]	"Chocolate"	"Cookie Dough"	"Mint Chocolate Chip"
##	[57,]	"Butter Pecan"	"Chocolate"	"Cookie Dough"
##	[58,]	"Chocolate"	"Coffee"	"Cookie Dough"
##	[59,]	"Chocolate"	"Mint Chocolate Chip"	"Mint Chocolate Chip"
##	[60,]	"Butter Pecan"	"Chocolate"	"Mint Chocolate Chip"
##	[61,]	"Chocolate"	"Coffee"	"Mint Chocolate Chip"
##	[62,]	"Butter Pecan"	"Butter Pecan"	"Chocolate"
##	[63,]	"Butter Pecan"	"Chocolate"	"Coffee"
##	[64,]	"Chocolate"	"Coffee"	"Coffee"
##	[65,]	"Strawberry"	"Strawberry"	"Strawberry"
##	[66,]	"Cookies & Cream"	"Strawberry"	"Strawberry"
##	[67,]	"Cookie Dough"	"Strawberry"	"Strawberry"
##	[68,]	"Mint Chocolate Chip"	"Strawberry"	"Strawberry"
##	[69,]	"Butter Pecan"	"Strawberry"	"Strawberry"
##	[70,]	"Coffee"	"Strawberry"	"Strawberry"
##	[71,]	"Cookies & Cream"	"Cookies & Cream"	"Strawberry"
##	[72,]	"Cookie Dough"	"Cookies & Cream"	"Strawberry"

```
## [73,] "Cookies & Cream" "Mint Chocolate Chip" "Strawberry"
## [74,] "Butter Pecan" "Cookies & Cream" "Strawberry"
## [75,] "Coffee" "Cookies & Cream" "Strawberry"
## [76,] "Cookie Dough" "Cookie Dough" "Strawberry"
## [77,] "Cookie Dough" "Mint Chocolate Chip" "Strawberry"
## [78,] "Butter Pecan" "Cookie Dough" "Strawberry"
## [79,] "Coffee" "Cookie Dough" "Strawberry"
## [80,] "Mint Chocolate Chip" "Mint Chocolate Chip" "Strawberry"
## [81,] "Butter Pecan" "Mint Chocolate Chip" "Strawberry"
## [82,] "Coffee" "Mint Chocolate Chip" "Strawberry"
## [83,] "Butter Pecan" "Butter Pecan" "Strawberry"
## [84,] "Butter Pecan" "Coffee" "Strawberry"
## [85,] "Coffee" "Coffee" "Strawberry"
## [86,] "Cookies & Cream" "Cookies & Cream" "Cookies & Cream"
## [87,] "Cookie Dough" "Cookies & Cream" "Cookies & Cream"
## [88,] "Cookies & Cream" "Cookies & Cream" "Mint Chocolate Chip"
## [89,] "Butter Pecan" "Cookies & Cream" "Cookies & Cream"
## [90,] "Coffee" "Cookies & Cream" "Cookies & Cream"
## [91,] "Cookie Dough" "Cookie Dough" "Cookies & Cream"
## [92,] "Cookie Dough" "Cookies & Cream" "Mint Chocolate Chip"
## [93,] "Butter Pecan" "Cookie Dough" "Cookies & Cream"
## [94,] "Coffee" "Cookie Dough" "Cookies & Cream"
## [95,] "Cookies & Cream" "Mint Chocolate Chip" "Mint Chocolate Chip"
## [96,] "Butter Pecan" "Cookies & Cream" "Mint Chocolate Chip"
## [97,] "Coffee" "Cookies & Cream" "Mint Chocolate Chip"
## [98,] "Butter Pecan" "Butter Pecan" "Cookies & Cream"
## [99,] "Butter Pecan" "Coffee" "Cookies & Cream"
## [100,] "Coffee" "Coffee" "Cookies & Cream"
## [101,] "Cookie Dough" "Cookie Dough" "Cookie Dough"
## [102,] "Cookie Dough" "Cookie Dough" "Mint Chocolate Chip"
## [103,] "Butter Pecan" "Cookie Dough" "Cookie Dough"
## [104,] "Coffee" "Cookie Dough" "Cookie Dough"
## [105,] "Cookie Dough" "Mint Chocolate Chip" "Mint Chocolate Chip"
## [106,] "Butter Pecan" "Cookie Dough" "Mint Chocolate Chip"
## [107,] "Coffee" "Cookie Dough" "Mint Chocolate Chip"
## [108,] "Butter Pecan" "Butter Pecan" "Cookie Dough"
## [109,] "Butter Pecan" "Coffee" "Cookie Dough"
## [110,] "Coffee" "Coffee" "Cookie Dough"
## [111,] "Mint Chocolate Chip" "Mint Chocolate Chip" "Mint Chocolate Chip"
## [112,] "Butter Pecan" "Mint Chocolate Chip" "Mint Chocolate Chip"
## [113,] "Coffee" "Mint Chocolate Chip" "Mint Chocolate Chip"
## [114,] "Butter Pecan" "Butter Pecan" "Mint Chocolate Chip"
## [115,] "Butter Pecan" "Coffee" "Mint Chocolate Chip"
## [116,] "Coffee" "Coffee" "Mint Chocolate Chip"
## [117,] "Butter Pecan" "Butter Pecan" "Butter Pecan"
## [118,] "Butter Pecan" "Butter Pecan" "Coffee"
## [119,] "Butter Pecan" "Coffee" "Coffee"
## [120,] "Coffee" "Coffee" "Coffee"
```

```
print(head(Scenario4, 10)) # First 10
```

```
##      [,1]      [,2]      [,3]
## [1,] "Vanilla" "Vanilla" "Vanilla"
## [2,] "Chocolate" "Vanilla" "Vanilla"
## [3,] "Strawberry" "Vanilla" "Vanilla"
```

```
## [4,] "Cookies & Cream"      "Vanilla"      "Vanilla"
## [5,] "Cookie Dough"        "Vanilla"      "Vanilla"
## [6,] "Mint Chocolate Chip" "Vanilla"      "Vanilla"
## [7,] "Butter Pecan"        "Vanilla"      "Vanilla"
## [8,] "Coffee"              "Vanilla"      "Vanilla"
## [9,] "Chocolate"           "Chocolate"     "Vanilla"
## [10,] "Chocolate"          "Strawberry"   "Vanilla"
```

```
print(tail(Scenario4, 10)) # Last 10
```

```
##           [,1]           [,2]           [,3]
## [111,] "Mint Chocolate Chip" "Mint Chocolate Chip" "Mint Chocolate Chip"
## [112,] "Butter Pecan"        "Mint Chocolate Chip" "Mint Chocolate Chip"
## [113,] "Coffee"              "Mint Chocolate Chip" "Mint Chocolate Chip"
## [114,] "Butter Pecan"        "Butter Pecan"        "Mint Chocolate Chip"
## [115,] "Butter Pecan"        "Coffee"               "Mint Chocolate Chip"
## [116,] "Coffee"              "Coffee"               "Mint Chocolate Chip"
## [117,] "Butter Pecan"        "Butter Pecan"        "Butter Pecan"
## [118,] "Butter Pecan"        "Butter Pecan"        "Coffee"
## [119,] "Butter Pecan"        "Coffee"               "Coffee"
## [120,] "Coffee"              "Coffee"               "Coffee"
```

```
print(Scenario4[sample(nrow(Scenario4), 5), ]) # 5 Random
```

```
##           [,1]           [,2]           [,3]
## [1,] "Mint Chocolate Chip" "Mint Chocolate Chip" "Strawberry"
## [2,] "Butter Pecan"        "Cookie Dough"        "Cookie Dough"
## [3,] "Cookies & Cream"      "Mint Chocolate Chip" "Vanilla"
## [4,] "Cookie Dough"        "Vanilla"              "Vanilla"
## [5,] "Butter Pecan"        "Cookies & Cream"      "Mint Chocolate Chip"
```

```
#120 total combos in first print
```

```
print(choose(8+3-1, 3)) #12c3 compare
```

```
## [1] 120
```

```
#same
```